Defying Gravity: Introduction to Circular Motion

SPH4U - Grade 12 Physics

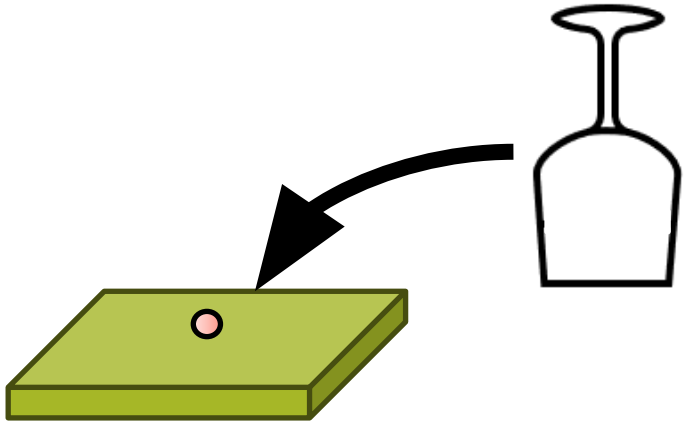
Mr. Dobson

Review

- **NEWTON'S FIRST LAW (INERTIA)**: Objects in motion stay in motion in a **STRAIGHT LINE unless acted upon** by an unbalanced force
- **FORCES:**
  - **Gravity (F_g):** pull downwards (mg)
  - **Normal Force (F_N):** Perpendicular to the surface, pushes away from the surface
  - **Net Force (F_{net}):** Unbalanced forces cause acceleration
- **FORCE COMPONENTS:**
  - Forces at angles split into Fx and Fy

Demo 1 – The Wine Glass Challenge

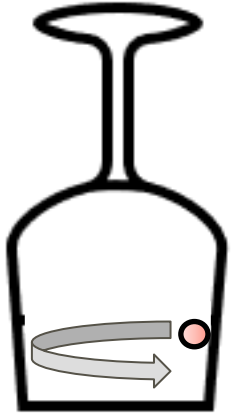
- **Setup**:
  - Wine glass
  - Marble
  - What happens when you spin and lift?



- **Predict**
  - When we spin the glass upside down and lift **will the marble fall out**?
- **Explain**
  - Why did you predict this? What can we use from previous lessons?

Observe Carefully!

- **Observe**
  - What did you observe? What happened to the marble?
- **Explain**
  - Thinking: What forces are acting?

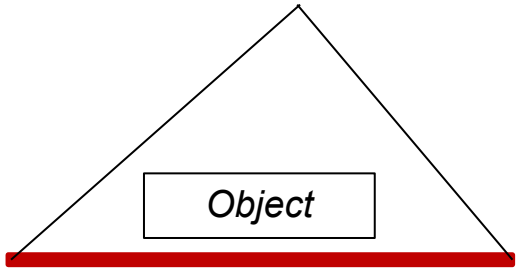


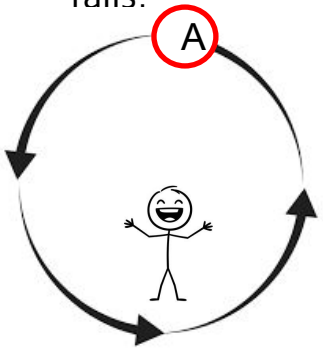
Demo 2 – The Spinning Board

- **Setup**
  - Board with an object on it
  - Swing it in a complete vertical circle
  - At the top, the board is **UPSIDE DOWN**

- **Prediction**
  - What do you think will happen when the board and object are at the top **(point A)?**

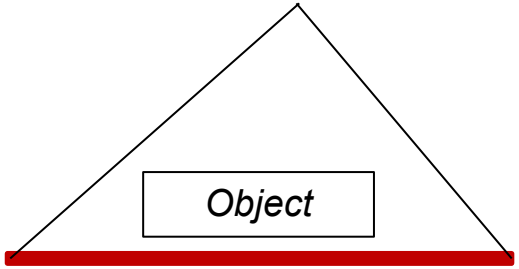
- **Explain**
  - What determines if it stays or falls?

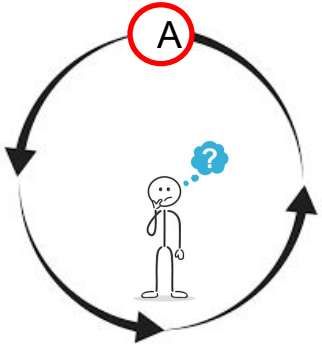




Observing *Carefully* (again)!

- **Observe**
  - What happened at the top (**point A**)?
- **Explain**
  - What forces might be acting on the object at the top (**A**)? the bottom?





Why it Works: The Physics Behind both Demos

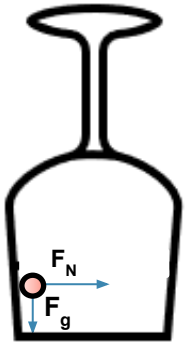
KEY IDEA: Circular motion requires a centre-seeking force (CENTRIPETAL FORCE)

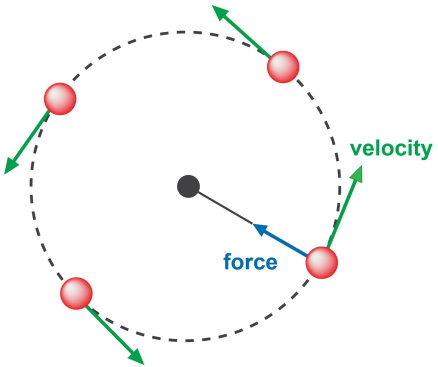
MARBLE IN GLASS

- Gravity pulls down
- Normal force redirects marble
- Fast spin
- Slow spin

BOARD IN VERTICAL CIRCLE

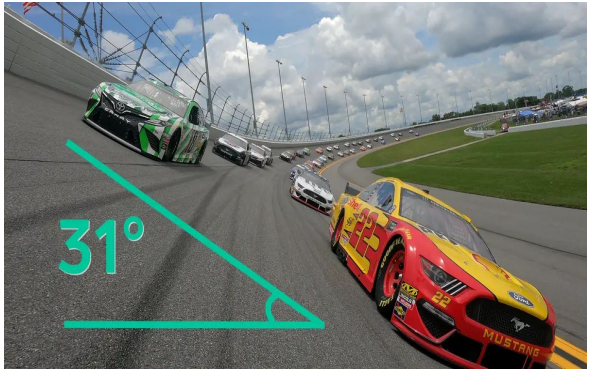
- Gravity pulls down
- Board pushes down (toward centre)
- Forces work together at the top
- Too slow → not enough inward force





Applications

- Based on what we saw, what are some similar applications of this concept?



NASCAR banked turns help keep drivers safe and faster

Olympic Gold Medalist in the Hammer Throw
Ethan Katzberg

Roller coaster

Learning Goal

We are learning …

- To analyze forces in uniform circular motion

Success Criteria

✓ I can identify the forces acting on an object moving in a circular path

✓ I can explain why objects can stay in circular motion (even when upside down!)

✓ I can apply force analysis to objects moving in vertical and horizontal circles

Today's Lesson

- **Next**: Circular motion free-body diagrams (FBD) and calculations!!

(END OF P.E.O.E)

Centrifuge

Washing Machine